Dalton Omens

Curriculum Vitae

San Francisco, CA
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Education

2020 - 2025 **Stanford University** Stanford, CA
Ph.D in Computer Science (SAIL)
Advisor: Ron Fedkiw

2016 - 2020 **University of California, Berkeley** Berkeley, CA
B.S. in Computer Science and Electrical Engineering
Summa Cum Laude (4.0)

Research Interests

AI for Virtual Characters and Worlds - How do we improve world engines, linear content generation, and virtual characters with AI tools? How do we ground multimodal generative AI for consistency, efficiency, and usefulness?

Democratizing Artistic Pipelines - Leveraging cutting-edge tech to enable traditionally prohibitive workflows such as character simulation/creation, content generation, emergent gameplay systems, and animation/retargeting.

Employment

2021 - Present **Epic Games**
World Models: Scene-level content generation informed by video diffusion models with a focus on game-readiness and semantic coherence.
AI characters in games: cognitive engines, memory systems, and automated evaluation using LLM, fine-tuned SLM, and NLI.
Virtual Humans: facial performance capture, natural language agents, and avatar creation using AI-informed optimization.

Summer 2023 **Sony Corporation of America**
Trained high quality physically-based neural simulations of loose clothing on human characters in conjunction with a small research team.

Summer 2020 **Unity Labs**
Worked with a research team to develop facial inverse rendering systems for animated characters using novel neural network techniques.

Summer 2019 **Blizzard Entertainment**
Used machine learning to predict render times and instantiating new pipelines for scientific computing in the Blizzard Animation studio.

Summer 2018 **NVIDIA**

Developed internal and production software for GeForce graphics card drivers.
Designed an internal application to analyze DisplayPort messages.

Publications

2026 **Improving Facial Rig Semantics for Tracking and Retargeting.** Dalton Omens, Allise Thurman, Jihun Yu, Ron Fedkiw. Computer Graphics Forum e70417. Presented at EUROGRAPHICS 2026, Aachen, Germany.

2024 **A Neural-Network-Based Approach for Loose-Fitting Clothing.** Yongxu Jin, Dalton Omens, Zhenglin Geng, Joseph Teran, Abishek Kumar, Kenji Tashiro, Ronald Fedkiw. arXiv, 2024.

2023 **Democratizing the Creation of Animatable Facial Avatars.** Yilin Zhu, Dalton Omens, Haodi He, Ron Fedkiw. arXiv, 2023.

2021 **Fast and Feature-Complete Differentiable Physics Engine for Articulated Rigid Bodies with Contact Constraints.** Keenon Werling, Dalton Omens, Jeongseok Lee, Ioannis Exarchos, C. Karen Liu. Robotics: Science and Systems (RSS), 2021.

2020 **Fast and Deep Facial Deformations.** Stephen W. Bailey, Dalton Omens, Paul Dilorenzo, and James F. O'Brien. ACM Transactions on Graphics, 39(4):94:1–15, August 2020. Presented at SIGGRAPH 2020, Washington D.C.

Teaching

2022 - 2025	CS 205L: Continuous Mathematical Methods with an Emphasis on Machine Learning	TA
2021 - 2024	CS 148: Introduction to Computer Graphics and Imaging	TA
2018 - 2020	CS 61A: The Structure and Interpretation of Computer Programs	uGSI
2017	CS 61A: The Structure and Interpretation of Computer Programs	Tutor
2017 - 2018	CS 198: Going Down the EECS Stack	Facilitator

Honors and Awards

2020	Stanford Reid Weaver Dennis Fellowship in the School of Engineering
2020	UC Berkeley Outstanding Graduate Student Instructor Award
2016 - 2019	UC Berkeley Dean’s List
2017 - 2020	IEEE Eta Kappa Nu EECS Honor Society
2016	BD Academic Achievement Scholarship